

Agent: Gas Leakage detection

Introduction: A Gas Leakage Detection and Automatic Shut-Off Agent is an automated safety system designed to detect gas leakage and take immediate action. When the detected gas level exceeds the defined safe limit, the agent identifies it as a gas leakage and automatically closes the main gas valve. At the same time, it activates an alarm and provides a warning message

Problem Statement: Gas leakage is a common household safety problem that may go unnoticed, especially when people are not present in the kitchen or nearby. Manually checking for leakage and turning off the gas supply can take time. Therefore, an automated system is needed that can continuously monitor gas levels, detect leakage, automatically shut off the main gas supply, and alert people immediately.

Problem Formulation: The problem can be formulated as an environment monitoring and decision-making problem. The agent receives gas-level readings from a gas sensor and compares them with a predefined safe limit.

Problem Solution: The proposed solution is an automated rule-based agent that continuously monitors gas sensor data.

The agent follows these steps:

Gas Sensor → Read Gas Level → Apply Rule → Detect Leakage → Shut Valve + Alarm + Warning.

Decision Rules:

IF gas level > safe limit

THEN

Gas Leakage = Detected

Gas Valve = OFF

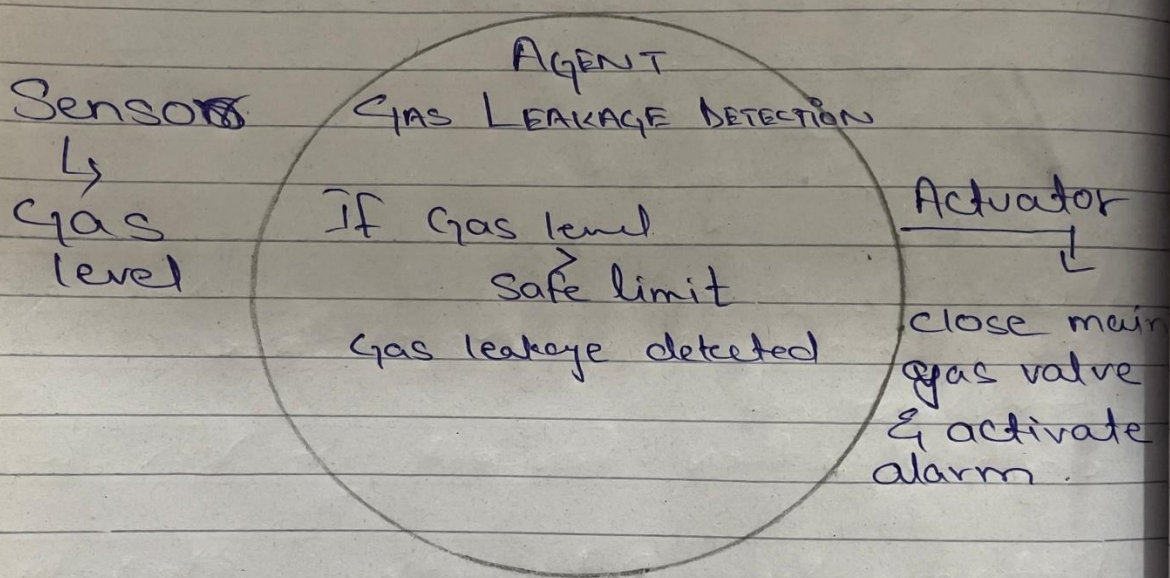
Alarm = ON

Warning Message = "Gas Leakage Detected!"

Else : Normal

Assignment

Agent : Gas leakage detection



Sensor :

- Gas sensor reading
- Temperature / Humidity
- Leakage status

Actuator :

Gas valve : OFF

Alarm : ON

PEAS Framework:

Performance: Detect leakage accurately, respond quickly

Environment: Kitchen

Actuators: Automatic gas shut-off valve

Sensors: Gas leakage sensor

ASSIGNMENT: **RULE BASE AGENT**

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COURSE: **ARTIFICIAL INTELLIGENCE**